

CLAUDE CERTIFIED ARCHITECT - FOUNDATIONS · EXAM CODE CCAR-F

Predict your score. Then prove it.

Many people rate themselves higher than they score. Write your practice-score guess below before you read on. Then answer five scenarios, sit a full timed mock, and compare the result with your guess.

[Sit a full timed mock →](#)[Or the 10-question diagnostic, no account needed → /practice/diagnostic](#)

Full mocks need a plan. Page 14 lists the four. The diagnostic needs no plan or account.

60

SCORED ITEMS

720

TO PASS, OF 1000

5

DOMAINS

120 min

EXAM TIME

12 mo

VALIDITY

[How every fact here is sourced → /evidence](#)

BEFORE YOU READ ON · YOUR FIRST PREDICTION

IF I SAT IT TODAY / 1000

MY TARGET / 1000

MY EXAM DATE

INSIDE THIS LEDGER

- 1 Your first prediction
- 2 Every core topic on one sheet
- 3 Find the constraint first
- 4 Tag every miss with one letter
- 5-9 Five scenarios, one per area
- 10-11 The answers, and your first gap
- 12 Ten-minute reviews
- 13 Review two mocks before you book
- 14 Pick how many mocks you need

NEXT · PAGE 2 [Every core topic on one sheet →](#)

The CCA-F exam on one sheet

The core topics, by area and weight. Tick the first box after you study a topic. Tick the second box only after you get its questions right on a timed test. The empty second boxes are what to study next.

Open this sheet online, every concept linked → /exam-guide

D1

Agentic Architecture & Orchestration

PAGE 5

27% · ≈16 ITEMS

STUDIED TESTED

Agentic Loops →	☐	☐
Subagents →	☐	☐
Stop Reason →	☐	☐
Session State & Persistence →	☐	☐
Human-in-the-Loop Escalation →	☐	☐
Subagent State Handoff →	☐	☐
Agent Instruction Files (CLAUDE.md / AGENTS.md) →	☐	☐
Task Decomposition Strategies →	☐	☐

D2

Tool Design & MCP Integration

PAGE 6

18% · ≈11 ITEMS

STUDIED TESTED

Tool Calling →	☐	☐
Tool Choice →	☐	☐
Model Context Protocol →	☐	☐
SDK Hooks (Pre/PostToolUse) →	☐	☐
Tool Evaluation & Testing →	☐	☐
Built-In Tools vs Custom Tools vs Skills vs MCP →	☐	☐

D3

Claude Code Configuration & Workflows

PAGE 7

20% · ≈12 ITEMS

STUDIED TESTED

CLAUDE.md Hierarchy →	☐	☐
Plan Mode →	☐	☐
Skills →	☐	☐
Checkpoints & Session Management →	☐	☐

D4

Prompt Engineering & Structured Output

PAGE 8

20% · ≈12 ITEMS

STUDIED TESTED

System Prompts & Instructions →	☐	☐
Prompt Caching →	☐	☐
Batch API →	☐	☐
Structured Outputs →	☐	☐
Vision & Multimodal →	☐	☐
Streaming →	☐	☐
Attention Engineering →	☐	☐
4D Framework →	☐	☐
Prompt Engineering Techniques →	☐	☐
Multi-Instance & Multi-Pass Review Architectures →	☐	☐

D5

Context Management & Reliability

PAGE 9

15% · ≈9 ITEMS

STUDIED TESTED

Context Window Management →	☐	☐
Case Facts Block →	☐	☐

NEXT · PAGE 3 Find the constraint first →

Find the constraint first

Every question has the same shape. Knowing the shape is worth a few marks on its own.

1 • The situation	A short description of a real setup and what is going wrong.
2 • The constraint	The phrase that rules options out: "without adding infrastructure", "lowest latency", "must be auditable". Find it first.
3 • The question	"Which is most effective..." asks for the best fit, not the only correct one.
4 • The options	Usually one right, one that does too much, one that patches a symptom, one that almost fits.

TRY IT · UNDERLINE THE CONSTRAINT, THEN PICK

A support agent must answer from the policy manual only, with no new services added. Answers sometimes cite rules that do not exist. What is the most effective fix?

- A • Fine-tune a model on the manual
- B • Put the manual in context and require a cited section for every answer
- C • Add a second model to check every answer
- D • Lower the temperature to 0

Answer: B. A adds work the constraint rules out. C is the bigger hammer. D patches a symptom.

More worked scenarios → /scenarios

Tag every miss with one letter

Tag every wrong answer with one letter. Pick the first reason that fits. Only K means you need new knowledge. The other four mean you knew it and lost the point anyway.

K

Didn't know it

"I had never seen this idea."

FIX

Read the concept page. Then explain the idea back in one sentence.

Concept pages → /concepts

C

Missed the constraint

"I knew it, but I skipped a word like only, first or without."

FIX

Underline every constraint in the scenario before you read the options.

Practise reading scenarios → /scenarios

H

Picked the bigger hammer

"I chose the strongest tool, not the one the scenario needs."

FIX

Ask: what is the smallest change that meets the need?

Common anti-patterns → /exam-guide/anti-patterns

S

Patched a symptom

"My answer hides the problem but leaves the cause."

FIX

Name the root cause in one line before you choose.

Common anti-patterns → /exam-guide/anti-patterns

T

Ran out of time

"I rushed or guessed near the end."

FIX

Practise to a clock. Our full mock gives 90 seconds per question.

Timed full mock → /pricing?cert=ccar-f

Agentic Architecture & Orchestration

1 · THE SCENARIO · FIND THE CONSTRAINT FIRST (PAGE 3)

SITUATION Your support agent runs a tool loop. Sometimes a reply says "All done!" and also holds a tool_use block. The loop exits on the words, and the refund never runs.

CONSTRAINT Fix it inside the loop you already have.

QUESTION What is the most reliable way to decide when the loop ends?

- ☐ A Look for words like "done" or "finished" in the reply text.
- ☐ B Cap the loop at 10 turns.
- ☐ C Branch on stop_reason: run the tool on "tool_use", exit on "end_turn".
- ☐ D Ask a second model to judge whether the task is finished.

2 · MY ANSWER

☐ A ☐ B ☐ C ☐ D

3 · HOW SURE AM I?

☐ Sure ☐ Unsure

THE ANSWER

On page 10, after all five

4 · THE CONSTRAINT, IN MY OWN WORDS

5 · THE OPTION I RULED OUT FIRST, AND WHY

NEXT · PAGE 6 Tool Design & MCP Integration →

Tool Design & MCP Integration

1 · THE SCENARIO · FIND THE CONSTRAINT FIRST (PAGE 3)

SITUATION A refund agent has three tools: `verify_customer`, `lookup_order` and `process_refund`. With `tool_choice` set to "any", it sometimes calls `lookup_order` first.

CONSTRAINT **Policy: identity is verified first, every time.**

QUESTION **Which change guarantees `verify_customer` runs first?**

- ☐ **A** Force it on the first turn: `tool_choice {"type": "tool", "name": "verify_customer"}`.
- ☐ **B** Keep "any" and add "always verify first" to the system prompt.
- ☐ **C** Remove `lookup_order` from the tool list.
- ☐ **D** Check each call afterwards and reject any made before verification.

2 · MY ANSWER

☐ A ☐ B ☐ C ☐ D

3 · HOW SURE AM I?

☐ Sure ☐ Unsure

THE ANSWER

On page 10, after all five

4 · THE CONSTRAINT, IN MY OWN WORDS

5 · THE OPTION I RULED OUT FIRST, AND WHY

Claude Code Configuration & Workflows

1 · THE SCENARIO · FIND THE CONSTRAINT FIRST (PAGE 3)

SITUATION Your team wants every Claude Code session in the repo to follow the same style and test rules. One engineer wrote the rules in ~/.claude/CLAUDE.md. Teammates' sessions ignore them.

CONSTRAINT The rules must be shared through git.

QUESTION Where should the rules live?

- ☐ A In each engineer's own ~/.claude/CLAUDE.md.
- ☐ B Pasted into the first prompt of every session.
- ☐ C In an MCP server that serves the rules to each session.
- ☐ D In the project CLAUDE.md in the repo, committed to git.

2 · MY ANSWER

☐ A ☐ B ☐ C ☐ D

3 · HOW SURE AM I?

☐ Sure ☐ Unsure

THE ANSWER

On page 10, after all five

4 · THE CONSTRAINT, IN MY OWN WORDS

5 · THE OPTION I RULED OUT FIRST, AND WHY

NEXT · PAGE 8 Prompt Engineering & Structured Output →

Prompt Engineering & Structured Output

1 · THE SCENARIO · FIND THE CONSTRAINT FIRST (PAGE 3)

SITUATION You ask Claude for JSON in the prompt. About 15% of replies add a sentence before the JSON, and your parser fails.

CONSTRAINT Production needs valid JSON every time.

QUESTION What is the most effective fix?

- ☐ A Add "Return JSON only, no other text" to the prompt, in capitals.
- ☐ B Define a tool whose input schema is the output, set strict: true, and force it with tool_choice.
- ☐ C Strip everything before the first { with a regex.
- ☐ D Fine-tune a model on clean JSON examples.

2 · MY ANSWER

☐ A ☐ B ☐ C ☐ D

3 · HOW SURE AM I?

☐ Sure ☐ Unsure

THE ANSWER

On page 10, after all five

4 · THE CONSTRAINT, IN MY OWN WORDS

5 · THE OPTION I RULED OUT FIRST, AND WHY

NEXT · PAGE 9 **Context Management & Reliability** →

Context Management & Reliability

1 · THE SCENARIO · FIND THE CONSTRAINT FIRST (PAGE 3)

SITUATION A support chat reaches turn 30. Older turns are summarised to save context. The agent now calls process_refund with amount "about \$250" instead of the exact order amount.

CONSTRAINT Keep summarising. Context is limited.

QUESTION What stops exact values from drifting?

- ☐ A Stop summarising and keep the full history.
- ☐ B Move to a model with a bigger context window.
- ☐ C Keep a case facts block with the exact values at the top of every prompt.
- ☐ D Ask the model to double-check numbers before each tool call.

2 · MY ANSWER

☐ A ☐ B ☐ C ☐ D

3 · HOW SURE AM I?

☐ Sure ☐ Unsure

THE ANSWER

On page 10, after all five

4 · THE CONSTRAINT, IN MY OWN WORDS

5 · THE OPTION I RULED OUT FIRST, AND WHY

NEXT · PAGE 10 The answers, and your first gap →

The answers, and your first gap

Mark each area right or wrong on its page. If you were sure and still wrong, that is your first gap. It is the mistake you would not have caught alone. None like that? Start with an unsure one. All five clear? Take the full mock.

RIGHT

____ of 5

SURE + WRONG

____ of 5

MY FIRST GAP IS

D ____

A score for every area: sit a full timed mock →

Needs a plan (page 14). Scored by area.

D1 Agentic Architecture

RIGHT: C

stop_reason is the field that says why Claude stopped. It is the loop's control signal.

A · S

This is the bug itself. Reply text can say "done" while a tool call is still waiting.

B · S

A cap hides the real signal. The loop still stops at the wrong time.

D · H

A second model is a heavier fix than needed. The answer is already in the response.

 Read next: [Stop Reason](#) → /concepts/stop-reason

D2 Tool Design & MCP

RIGHT: A

Only a named tool_choice forces one specific tool. "any" lets the model pick which tool.

B · C

A prompt makes it likely, not certain. The constraint says every time.

C · H

It breaks the rest of the flow, which still needs lookup_order.

D · S

It catches the wrong order after it happens instead of preventing it.

 Read next: [Tool Choice](#) → /concepts/tool-choice

D3 Claude Code

RIGHT: D

Project-level CLAUDE.md loads at session start for everyone who clones the repo.

A · C

User-level files are personal. Git never shares them.

B · S

It works for one session and is gone in the next.

C · H

A new server for what a committed file already does.

 Read next: [CLAUDE.md Hierarchy](#) → /concepts/claude-md-hierarchy

The answers, continued

D4
Prompt Engineering

RIGHT: B

A strict, forced tool call can only return input that matches the schema.

A · C

Prompts are likely, not certain. The constraint says every time.

C · S

It hides the problem and breaks on the next odd reply.

D · H

Much more work, and still no guarantee.

Read next: Structured Outputs → /concepts/structured-outputs

D5
Context Management

RIGHT: C

Exact facts live outside the summarised history, so they never get paraphrased.

A · C

It breaks the constraint. Summaries still stay.

B · H

A bigger window delays the problem. Summaries still lose precision.

D · S

The model checks against a summary that has already lost the number.

Read next: Case Facts Block → /concepts/case-facts-block

FIVE QUESTIONS ARE A SAMPLE · A FULL MOCK GIVES YOU

60 questions at 90 seconds each, a tighter clock than the real exam. You get a score out of 1000, a score for each area, and every miss linked to its concept page.

Ten minutes a day, from memory

Pick one area a day. Close your notes, answer the prompt, then check the concept page. A wrong answer here costs nothing. It tells you what to read before the mock.

Check all five areas in one full mock →

Needs a plan (page 14). Sit it after the five reviews.

ONE PROMPT PER AREA · ANSWER FROM MEMORY, THEN CHECK

- D1

Agentic Architecture
When should an agent stop its loop, and how does it know? When does a task need a subagent?
Check: Subagents → /concepts/subagents

☐ GOT IT
☐ SHAKY
☐ MISSED
-
- D2

Tool Design & MCP
When do you force a tool call, and when do you let the model choose? What does MCP add over a built-in tool?
Check: Model Context Protocol → /concepts/mcp

☐ GOT IT
☐ SHAKY
☐ MISSED
-
- D3

Claude Code
Which CLAUDE.md file wins when two give different rules? When is plan mode the right first step?
Check: Plan Mode → /concepts/plan-mode

☐ GOT IT
☐ SHAKY
☐ MISSED
-
- D4

Prompt Engineering
How do you make output always match a schema? What belongs in the system prompt, not the user turn?
Check: System Prompts & Instructions → /concepts/system-prompts

☐ GOT IT
☐ SHAKY
☐ MISSED
-
- D5

Context Management
The context is nearly full. What do you keep, and what goes into a case facts block?
Check: Context Window Management → /concepts/context-window

☐ GOT IT
☐ SHAKY
☐ MISSED
-

Review two mocks before you book

This is the only score sheet. Fill one column at a time: your guess, the scenarios, then two full mocks. Read each row across. The weakest row is what to study next.

THE LEDGER · ONE COLUMN AT A TIME

AREA	WEIGHT	MY GUESS	SCENARIO	MOCK 1	MOCK 2	LETTER
D1 Agentic Architecture	27%					
D2 Tool Design & MCP	18%					
D3 Claude Code	20%					
D4 Prompt Engineering	20%					
D5 Context Management	15%					
Overall	/ 1000					

THE BOOKING RULE · USE THE TARGET FROM PAGE 1

IF MOCK 2 IS...	THEN
At or above your target, and no area is weak	Book by your own rule. A mock score is practice, not a prediction.
Up from mock 1, but below your target	Add a week. Fix your new top three, then sit another mock.
Flat or down from mock 1	Go back to page 4. Fix the letter you repeat most.

720 is the real exam's pass mark. Set your target with some room above it.

WRITE DOWN YOUR PRACTICE DATES

MOCK 1 · MY DATE

MOCK 2 · MY DATE

MY EXAM · BOOKED FOR

Pick how many mocks you need

The booking rule on page 13 needs two full mocks. A mock is a safe place to be wrong. It never counts toward the real exam.

[Start mock 1 with Pro →](#)
[Compare plans → /pricing](#)

Pro gives three full mocks per certification. That covers mock 1, mock 2, and a spare if the rule asks for an extra week.

THE FOUR PLANS · WHAT EACH ONE COVERS

Single

One full mock, on one certification you pick.

[See the Single mock →](#)

</checkout?plan=single&cert=ccar-f>

FOR THIS LEDGER

Mock 1 only

Base

One full mock for each of the four certifications.

[See Base →](#)

</checkout?plan=base&cert=ccar-f>

FOR THIS LEDGER

Mock 1 only

Pro

Three full mocks per certification, plus every study tool: warm-ups, an adaptive study plan, a readiness meter, progress emails and flashcards.

[See Pro →](#)

</checkout?plan=pro&cert=ccar-f>

FOR THIS LEDGER

Mock 1, mock 2 and a spare

All-Access

Unlimited full mocks on all four certifications, plus every study tool.

[See All-Access →](#)

</checkout?plan=all-access&cert=ccar-f>

FOR THIS LEDGER

As many as you need

EVERY PLAN

90 DAYS

Access lasts 90 days.

ONE PAYMENT

No auto-renew. Nothing is charged again.

MOVE UP LATER

While a plan is active, pay only the difference. The end date stays the same.

[Exam day checklist → /exam-guide/day-of](#)

[Registration and cost → /exam-guide/registration-and-cost](#)

Good luck with the exam.

CAC Editorial